Supplementary Appendices

An analytical cyclo-steady-state model for inter-rifting deformation at magmatic rifts

A Fourier reduction and the characteristic velocity

Equation (1) (the governing PDE) is non-local in x and not tractable in physical space. A spatial Fourier transform resolves this, because the Stokes convolution diagonalizes in wavenumber space. Defining the transform pair

$$\hat{u}(k,t) = \int_{-\infty}^{+\infty} u(x,t) e^{-ikx}\, dx, \qquad u(x,t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \hat{u}(k,t) e^{ikx}\, dk,$$

the transform of $\partial^2 u/\partial x^2$ is $-k^2\hat{u}$. The transform of the Stokes kernel convolution is the classical result (Savage and Prescott, 1978):

$$\mathcal{F}\left\{ \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2} \frac{\partial u}{\partial t}\, dx' \right\} = -2\eta|k| \frac{\partial \hat{u}}{\partial t}. \tag{1}$$

Applying the Fourier transform to the governing PDE and using (1):

$$-\frac{EH}{1-\nu^2} k^2 \hat{u} - 2\eta|k| \frac{\partial \hat{u}}{\partial t} = 0. \tag{2}$$

Rearranging and introducing the characteristic velocity

$$a := \frac{EH}{2\eta(1-\nu^2)}, \tag{3}$$

equation (2) simplifies to a decoupled first-order ODE for each wavenumber k:

$$\frac{\partial \hat{u}}{\partial t} = -a\,|k|\,\hat{u}. \tag{4}$$

The parameter a encodes the full elastic–viscous coupling of the system: large a (stiff crust, low viscosity) implies rapid redistribution of post-diking strain over broad spatial scales; small a implies slow, locally confined relaxation. In physical space, the operator $|k|$ corresponds to the fractional half-Laplacian $(-\Delta)^{1/2}$, so (4) is equivalently written as the fractional diffusion equation (Metzler and Klafter, 2000; Samko and Ross, 1993; Samko et al., 1993)

$$\frac{\partial u}{\partial t} = -a\,(-\Delta)^{1/2} u. \tag{5}$$

This contrasts with classical Fickian diffusion: the half-order operator, a direct consequence of the non-local Stokes kernel, produces Lorentzian rather than Gaussian spatial profiles, with tails that decay as $1/x$ rather than exponentially. The solution of (4) for arbitrary initial data is:

$$\hat{u}(k,t) = \hat{u}(k,0)\,e^{-a|k|t}. \tag{6}$$

B Step response: Sokhotski–Plemelj and Dirichlet evaluation

A symmetric opening of 2ω at $t = 0$ displaces the right side of the rift by $+\omega$ and the left by $-\omega$. Exploiting the antisymmetry, the initial condition is a Heaviside displacement:

$$u(x, 0^+) = \omega\, \mathcal{H}(-x). \tag{7}$$

Applying the Fourier transform to (7) gives $\hat{u}(k, 0) = \omega \int_{-\infty}^{0} e^{-ikx}\, dx$, which for real k is treated in the distributional sense by adding a convergence regulator $\varepsilon > 0$:

$$\int_{-\infty}^{0} e^{-ikx}\, dx = \lim_{\varepsilon \to 0} \int_{-\infty}^{0} e^{(-ik+\varepsilon)x}\, dx = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon - ik}.$$

The Sokhotski–Plemelj theorem gives $\pi\delta(k) + i/k$ in the limit, so the initial Fourier spectrum is:

$$\hat{u}(k, 0) = \omega\pi\delta(k) + \frac{i\omega}{k}. \tag{8}$$

Substituting (8) into (6) and noting that $\delta(k)$ is active only where $e^{-a|k|t} = 1$:

$$\hat{u}(k, t) = \omega\pi\delta(k) + \frac{i\omega}{k}\, e^{-a|k|t}. \tag{9}$$

The inverse Fourier transform splits into two contributions, $u(x, t) = I_1 + I_2$. The delta-function term gives a constant background $I_1 = \omega/2$; for I_2, the parity of $\sin(kx)/k$ yields:

$$I_2 = -\frac{\omega}{\pi} \int_0^{\infty} \frac{\sin(kx)}{k}\, e^{-akt}\, dk. \tag{10}$$

This is a standard Dirichlet-type integral. Applying the complementary arctangent identity $\arctan(z) + \arctan(1/z) = \pi/2$, the step displacement is:

$$u^{\mathrm{STEP}}(x,t) = \frac{\omega}{\pi} \arctan\left(\frac{at}{x}\right). \tag{11}$$

Differentiating with respect to t, the step velocity is a Lorentzian profile that broadens in time:

$$v^{\mathrm{STEP}}(x,t) = \frac{\omega a}{\pi} \frac{x}{x^2 + a^2 t^2}. \tag{12}$$

Both expressions satisfy the correct limits: $u^{\mathrm{STEP}} \to \omega/2$ and $v^{\mathrm{STEP}} \to 0$ as $t \to \infty$, consistent with far-field plate divergence. The velocity peak migrates outward as at, with amplitude decaying as $1/t$.

C CSS Digamma representation

The CSS velocity arises by extending the finite sum over past intrusions to an infinite history. Introducing $\tau \in [0, T)$ as the time elapsed since the most recent intrusion:

$$v_{\mathrm{CSS}}(x,\tau) = \frac{\omega a}{\pi} \sum_{j=1}^{\infty} \frac{x}{x^2 + a^2(\tau - jT)^2}. \tag{13}$$

Because this series converges absolutely, it admits a closed-form expression via the Digamma function $\psi(z)$. Using the partial-fraction identity

$$\frac{x}{x^2 + a^2(\tau - jT)^2} = \frac{1}{2i}\left[\frac{1}{a(\tau - jT) - ix} - \frac{1}{a(\tau - jT) + ix}\right],$$

factoring $-aT$ from each denominator and substituting into (13), the double sum takes the form of differences of Digamma series. Using the representation $\psi(z+1) = -\gamma + \sum_{n=1}^{\infty}(1/n - 1/(n+z))$, the Euler–Mascheroni constant γ and the $1/n$ terms cancel when the two conjugate series are differenced. With $Z = \tau/T + ix/(aT)$ and the identity

$\psi(Z) - \psi(\bar{Z}) = 2i\,\mathrm{Im}\{\psi(Z)\}$, the CSS velocity collapses to:

$$v_{\mathrm{CSS}}(x,\tau) = \frac{2\omega}{\pi T}\,\mathrm{Im}\Big\{\psi\Big(\frac{\tau}{T} + i\frac{x}{aT}\Big)\Big\}\,. \tag{14}$$

Integrating over τ, and using the antiderivative $d[\ln\Gamma(z)]/dz = \psi(z)$, the CSS displacement is:

$$u_{\mathrm{CSS}}(x,\tau) = \frac{2\omega}{\pi}\,\mathrm{Im}\Big\{\ln\Gamma\Big(\frac{\tau}{T} + i\frac{x}{aT}\Big) - \ln\Gamma\Big(i\frac{x}{aT}\Big)\Big\}\,. \tag{15}$$

The far-field conservation property $\int_0^T v_{\mathrm{CSS}}\,d\tau = \omega$ follows from $\Gamma(1+iq) = iq\,\Gamma(iq)$, which implies $\mathrm{Im}\{\ln\Gamma(1+iq) - \ln\Gamma(iq)\} = \pi/2$, giving the integral ω.

D Maxwell visco-elastic kernel and governing equation

Elastic Stokes kernel and the correspondence principle

For a purely elastic half-space the basal traction relates to surface displacement through a kernel proportional to μ_m:

$$\sigma_{xz}^{\mathrm{BOT,\,el}}(x) = \int_{-\infty}^{+\infty} \frac{2\mu_m}{\pi(x-x')^2}\,u(x')\,dx'. \tag{16}$$

The Maxwell constitutive relation for deviatoric stress τ_{ij} and strain ε is:

$$\dot{\varepsilon} = \frac{\dot{\tau}}{2\mu_m} + \frac{\tau}{2\eta}. \tag{17}$$

To adapt (16) we employ the elastic–visco-elastic correspondence principle (Hetland and Hager, 2005; Barbot and Fialko, 2010): in the Laplace domain, static moduli are replaced by their complex, frequency-dependent counterparts. Applying the Laplace

transform $\tilde{f}(s) = \mathcal{L}\{f(t)\}$ with zero initial conditions gives:

$$s\tilde{\varepsilon} = \tilde{\tau}\left(\frac{s}{2\mu_m} + \frac{1}{2\eta}\right). \tag{18}$$

Solving for stress gives $\tilde{\tau} = 2\tilde{\mu}(s)\tilde{\varepsilon}$, where the effective frequency-dependent shear modulus of the Maxwell body is:

$$\tilde{\mu}(s) = \frac{\mu_m \eta\, s}{\mu_m + \eta s} = \mu_m \frac{\tau_R s}{1 + \tau_R s}, \tag{19}$$

with the Maxwell relaxation time $\tau_R := \eta/\mu_m$.

Visco-elastic Stokes kernel in time domain

Applying the correspondence principle (replacing μ_m with $\tilde{\mu}(s)$ in the Laplace domain) and factoring out $\eta = \mu_m \tau_R$:

$$(1 + \tau_R s)\, \tilde{\sigma}_{xz}^{\mathrm{BOT}}(x, s) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x - x')^2} \left[s\tilde{u}(x', s)\right] dx'. \tag{20}$$

Inverting to the time domain (multiplication by s in the Laplace domain corresponds to $\partial/\partial t$) yields the visco-elastic Stokes kernel:

$$\sigma_{xz}^{\mathrm{BOT}} + \tau_R \frac{\partial \sigma_{xz}^{\mathrm{BOT}}}{\partial t} = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x - x')^2} \frac{\partial u(x', t)}{\partial t}\, dx'. \tag{21}$$

In the limit $\tau_R \to 0$, (21) reduces to the purely viscous Stokes kernel.

Governing equation in Fourier space

The equilibrium equation expresses the basal traction in terms of the elastic crust displacement:

$$\sigma_{xz}^{\mathrm{BOT}} = -\frac{EH}{1 - \nu^2} \frac{\partial^2 u}{\partial x^2}. \tag{22}$$

Substituting this expression and its time derivative into (21), then applying the spatial Fourier transform using (1):

$$\frac{EH}{1-\nu^2}k^2\hat{u} + \tau_R\frac{EH}{1-\nu^2}k^2\frac{\partial\hat{u}}{\partial t} = -2\eta|k|\frac{\partial\hat{u}}{\partial t}. \tag{23}$$

Dividing by 2η, using (3), and collecting $\partial\hat{u}/\partial t$ terms gives the visco-elastic analogue of (4):

$$\frac{\partial\hat{u}}{\partial t} = -\left(\frac{a|k|}{1+\tau_R a|k|}\right)\hat{u}. \tag{24}$$

The denominator $1+\tau_R a|k|$ acts as a low-pass filter on the relaxation rate, slowing the decay of high-wavenumber (near-field) features relative to the purely viscous case.

Initial condition evolution and step response

Equation (24) is separable in k, yielding a decaying exponential for each wavenumber:

$$\hat{u}(k,t) = \hat{u}(k,0)\exp\left(-\frac{a|k|}{1+\tau_R a|k|}\,t\right). \tag{25}$$

Applying the Heaviside initial condition (8):

$$\hat{u}(k,t) = \omega\pi\delta(k) + \frac{i\omega}{k}\exp\left(-\frac{a|k|}{1+\tau_R a|k|}\,t\right). \tag{26}$$

Taking the inverse Fourier transform and following the same parity argument as in Appendix B, the Maxwell step displacement is:

$$u^{\mathrm{MAX}}(x,t) = \frac{\omega}{2} - \frac{\omega}{\pi}\int_0^{+\infty}\frac{\sin(kx)}{k}\exp\left(-\frac{akt}{1+\tau_R ak}\right)dk, \tag{27}$$

since the rational exponent prevents the closed-form Dirichlet evaluation of Appendix B. The surface velocity follows by differentiating with respect to t:

$$v^{\mathrm{MAX}}(x,t) = \frac{\omega a}{\pi} \int_0^{+\infty} \frac{\sin(kx)}{1+\tau_R a k} \exp\left(-\frac{akt}{1+\tau_R a k}\right) dk. \tag{28}$$

The exponent reveals two physically distinct regimes. In the long-wavelength limit ($k \to 0$ or $t \to \infty$), $1+\tau_R a k \approx 1$ and the exponential reduces to $\exp(-akt)$, recovering the purely viscous Dirichlet integral of Appendix B. In the short-wavelength limit ($k \to \infty$ or $t \to 0$), $1+\tau_R a k \approx \tau_R a k$ and the exponent saturates:

$$\lim_{k\to\infty} \left(-\frac{akt}{1+\tau_R a k}\right) = -\frac{t}{\tau_R}. \tag{29}$$

At early times $t \ll \tau_R$, the exponential approaches unity and no near-field deformation occurs outside the rift interface. For $t \sim \tau_R$, near-field deformation decays exponentially at the fixed rate $1/\tau_R$, representing pure Maxwell relaxation independent of the elastic crust.

E Maxwell Cyclo-Steady State: geometric series and periodicity condition

The Maxwell CSS is the superposition of step-response velocities from all preceding intrusions:

$$v^{\mathrm{MAX}}_{\mathrm{CSS}}(x,\tau) = \sum_{j=0}^{\infty} v^{\mathrm{MAX}}(x,\, \tau + jT). \tag{30}$$

Writing v^{MAX} in Fourier space as $\frac{2\omega a}{\pi} \int_0^{\infty} \sin(kx)\, e^{-\alpha(k)(\tau+jT)}\, dk$, with the effective Maxwell decay rate

$$\alpha(k) := \frac{ak}{1+\tau_R a k}, \tag{31}$$

each past intrusion contributes a factor $e^{-\alpha(k)jT}$. The series converges for all $k > 0$ and the geometric sum $\sum_{j=0}^{\infty} e^{-\alpha(k)jT} = (1 - e^{-\alpha(k)T})^{-1}$ can be performed inside the Fourier integral. The same result follows directly from the requirement that each Fourier mode satisfy the periodicity condition $\hat{u}(k, \tau + T) = \hat{u}(k, \tau) + i\omega/k$, which admits the unique solution

$$\hat{u}(k, \tau) = \frac{i\omega/k}{1 - e^{-\alpha(k)T}} \, e^{-\alpha(k)\tau}, \qquad \tau \in [0, T). \tag{32}$$

Inverting gives the Maxwell CSS velocity and displacement:

$$v_{\mathrm{CSS}}^{\mathrm{MAX}}(x, \tau) = \frac{2\omega a}{\pi} \int_0^{\infty} \frac{\sin(kx)}{1 + \tau_R a k} \, \frac{e^{-\alpha(k)\tau}}{1 - e^{-\alpha(k)T}} \, dk, \tag{33}$$

$$u_{\mathrm{CSS}}^{\mathrm{MAX}}(x, \tau) = \frac{2\omega}{\pi} \int_0^{\infty} \frac{\sin(kx)}{k} \, \frac{1 - e^{-\alpha(k)\tau}}{1 - e^{-\alpha(k)T}} \, dk. \tag{34}$$

Integrating the velocity over one full period, using $\int_0^T e^{-\alpha\tau}/(1 - e^{-\alpha T}) \, d\tau = 1/\alpha$ and the algebraic identity $a/[(1 + \tau_R a k) \, \alpha(k)] = 1/k$, gives

$$\int_0^T v_{\mathrm{CSS}}^{\mathrm{MAX}}(x, \tau) \, d\tau = \frac{2\omega}{\pi} \int_0^{\infty} \frac{\sin(kx)}{k} \, dk = \omega, \tag{35}$$

where the last step is the Dirichlet integral. The full rift half-opening is recovered every cycle, independent of τ_R. In the limit $\tau_R \to 0$, $\alpha(k) \to ak$ and $(1 + \tau_R a k)^{-1} \to 1$, and the Maxwell CSS velocity reduces to the viscous CSS Fourier integral whose closed form is the Digamma series of Appendix C.